

BHAVANESH NIDAMANURI

Machine Learning & AI Engineer

Seeking a 6-month internship (stage) in AI / Machine Learning | Available from January 2027

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Profile

Master's student in Artificial Intelligence at EPITA with over a year of hands-on experience in data analysis and machine learning. I build end-to-end AI systems, from data preparation and modeling to API deployment, including a full-stack RAG application with Gemini, audio deep learning and MLOps projects.

Education

EPITA – School of Engineering and Computer Science

Master's in Computer Science – Artificial Intelligence

Paris, France

2025 – Present

RGUKT Ongole

Bachelor's in Computer Science

Andhra Pradesh, India

2020 – 2024

Projects

Enterprise AI Knowledge Assistant – RAG Platform

Stack: Python, FastAPI, PostgreSQL, ChromaDB, Sentence Transformers, Google Gemini, JWT

- Built an end-to-end RAG application enabling authenticated users to upload PDF, DOCX and TXT documents and query them in natural language.
- Developed the full pipeline: text extraction, recursive chunking, 384-dimensional embeddings (Sentence Transformers), ChromaDB vector search with user-specific retrieval, and source-grounded Gemini answers.
- Secured the API with JWT authentication, bcrypt password hashing and per-user document ownership, with persistent storage in PostgreSQL.

Audio Forensic Pipeline for Deepfake Detection and Content Verification

Stack: Python, CNN + BiLSTM, Grad-CAM, Speech-to-Text

- Designed an end-to-end forensic pipeline with a bimodal CNN + BiLSTM model to detect audio deepfakes.
- Localized suspicious segments (Grad-CAM, saliency maps, temporal analysis) and added speech-to-text with Wikipedia-based fact verification to produce explainable forensic reports.

Gym Heart Attack Risk Prediction – ML & MLOps Pipeline

Stack: Python, Scikit-learn, FastAPI, Streamlit, Airflow, Docker Compose, Grafana

- Built Logistic Regression and Random Forest classifiers (92.4% accuracy); feature engineering added +28%.
- Served predictions through a FastAPI backend (latency under 200 ms) with a Streamlit web app where gym members enter their data and get a risk prediction.
- Automated data ingestion, processing and prediction workflows with Airflow DAGs, monitored system metrics with Grafana dashboards, and containerized all services with Docker Compose.

Experience

Cognizant Technology Solutions

Data Analyst

Hyderabad, India

May 2024 – Feb 2025

- Analyzed business data to support decision-making and built Power BI dashboards to track KPIs.
- Optimized SQL/ETL data pipelines to improve operational efficiency and reporting.

Oasis Infobyte

Data Science Intern

Hyderabad, India

Aug 2023 – Jan 2024

- Built machine learning models (Logistic Regression, Random Forest) reaching 85%+ accuracy on real-world data.
- Performed EDA and feature engineering on 10,000+ records, reducing model error by 15%, and delivered an end-to-end predictive pipeline (Python, Scikit-learn).

Technical Skills

- Languages:** Python, SQL, Java, JavaScript, C
- ML & Deep Learning:** Scikit-learn, PyTorch, TensorFlow, Keras, XGBoost, LightGBM, CNN, LSTM, Feature Engineering, Model Evaluation
- GenAI & NLP:** RAG, Google Gemini, OpenAI API, LangChain, Embeddings, Semantic Search, ChromaDB, Hugging Face, Prompt Engineering, Speech-to-Text
- Backend & MLOps:** FastAPI, Streamlit, REST APIs, JWT Auth, SQLAlchemy, Docker, Docker Compose, Airflow, Grafana, Git/GitHub
- Data & Cloud:** Pandas, NumPy, Power BI, PostgreSQL, MySQL, MongoDB, ETL, GCP (BigQuery, Cloud Storage), AWS (S3, EC2)

Languages & Soft Skills

- Spoken languages:** English (C2), French (B1), Telugu (C2), Hindi (B2)
- Soft skills:** Problem-solving, Analytical thinking, Teamwork, Communication, Adaptability, Fast learner